Future Engineers

Sojourner

PRV Robotics

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Team Presentation

The team consists of three members: Eliel Salas, Aurelio Perez, and Yeriel Cruz, who are 17, 17, and 15 years old, respectively, and are currently in 11th, 11th, and 10th grade. Each member has specific roles that contribute to the development of the project. Eliel is responsible for managing the YouTube content, supervising programming and construction, taking notes, and documenting the project. Aurelio is in charge of field operations, ensuring the robot performs effectively during testing and competition. Yeriel serves as the master builder and programmer, leading the construction of the robot and the development of its control software.

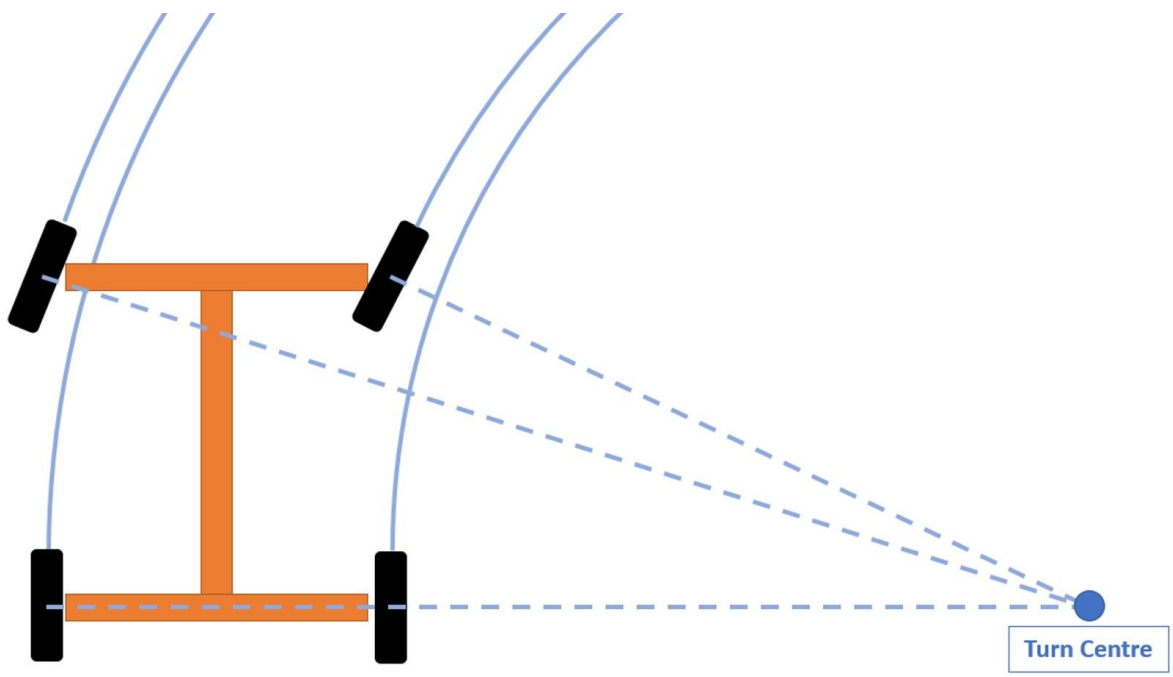
Project Overview

Our robot is an obstacle dodger that is designed to accomplish certain challenges under the rules of institution, World Robot Olympiad, specifically in the Future Engineers category. The idea is to be able to navigate through a course on your own, and to sense and avoid obstacles as they come up. The system includes sensors, programming logic, and mechanical design to ensure accuracy and efficiency during operation. The robot follows the competition rules and shows both innovation and adherence to engineering standards. This project is ultimately a demonstration of how we can take our problem-solving skills and technological knowledge and apply it in a competitive robotics environment.

Our robot is equipped with car-like locomotive functions, so it can move along the course efficiently. It uses two sensors to distinguish between colors and help it find its way, detecting obstacles and finding the right path. For the components, we added the two motors to the front and back of the robot. These motors work together with precision to move the wheels of the vehicle forward and backwards. This combination of sensors and motors enables the robot to move on its own while still being controlled and stable.

Design Process & Initial Concepts

We started with an idea to build a robot with two wheels and a ball bearing that can move forward and backward. This robot was easy to make and program. We did not like this though because it did not have a way to steer like a car. The two-wheeled robot was also not stable when it made turns on the track. Then we thought about adding parts like distance and color sensors to the robot. These sensors would help the robot know what is around it. In the end, we built a robot with four wheels, a color sensor, and a distance one. This design was good because it was more stable and we had more control over the robot. It could follow the track and react to things in its way.

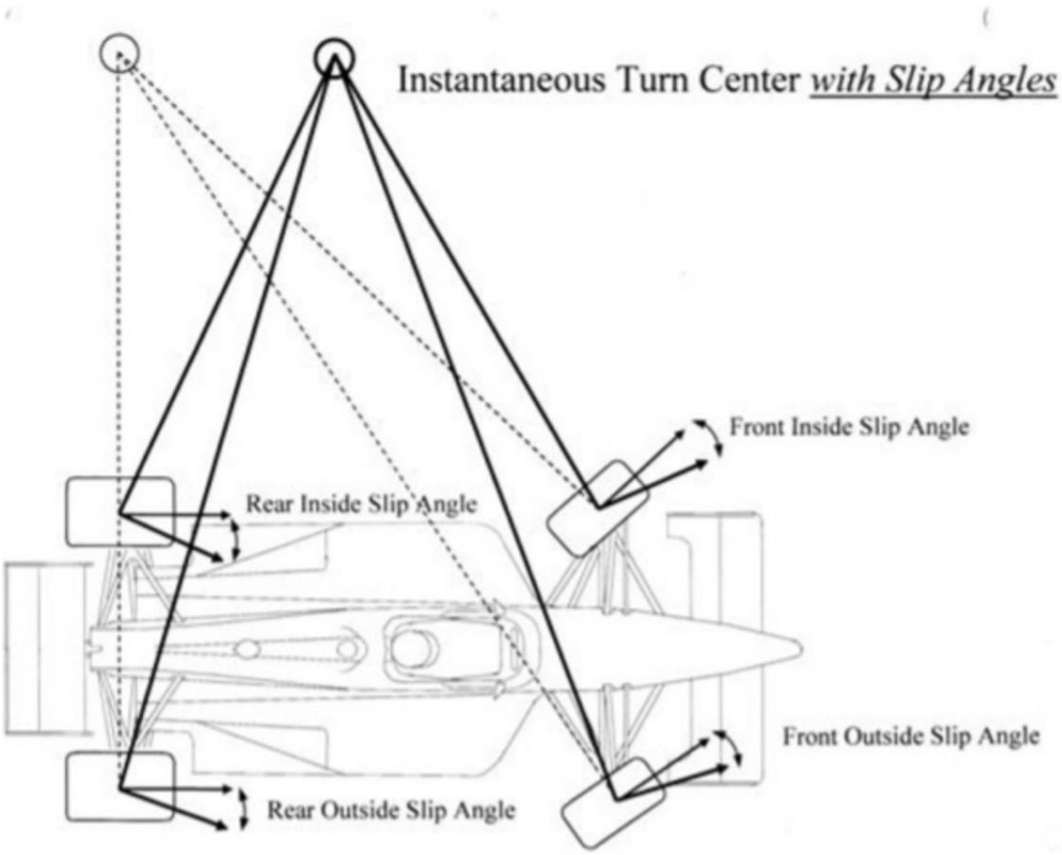


System Overview

The robot is composed of four main subsystems: mechanical structure, sensing system, power system, and software control. The mechanical system consists of a four-wheel chassis with stability during movements on the track. Steering system that allows for directional control like a real vehicle and propulsion by two motors. The sensing system is based on color sensors for detecting track markings and traffic signs. Then you also have sensors that feed real time data to the controller, so the robot knows where it is and can adjust its path. The power system consists of a rechargeable battery that provides energy to all electronic components (motors and sensors). The software system receives sensor input and decides how to control movement. It incorporates obstacle detection, path correction, and speed control so that the robot can efficiently complete the course.

Design Tradeoffs

The choice of control systems was a major design decision. The team evaluated both the LEGO Spike Prime and EV3 platforms. Spike was selected in our evaluation, as it provided more reliable performance and greater precision for motor control. The second key tradeoff was speed versus accuracy. Going faster could mean finishing sooner but also raising the possibility of losing control or running into obstacles. The team concentrated on accuracy and stability to perform the same in several runs. Also, the wheel mounting system was designed with turntable elements and not direct mount. This improved structural stability and reduced the chance of mechanical failure during operation.



Components & Hardware

The robot is built using a combination of mechanical and electronic components that work together to enable autonomous navigation and obstacle avoidance.

- Motors:
  The robot uses motors from the LEGO Spike system to provide movement. These motors are responsible for driving the wheels forward and backward, allowing the robot to move efficiently through the course. Their precision enables smoother acceleration and better control when navigating obstacles.
- Sensors:
  The robot is equipped with color and ultrasonic sensors. The color sensors are used to detect track markings and traffic signs, helping the robot maintain its path and make navigation decisions. The ultrasonic sensor is used to measure distance to nearby objects, assisting in obstacle detection and avoidance. Together, these sensors provide the data needed for consistent and accurate movement.
- Controller:
  The robot is controlled by a LEGO Spike Prime hub, which acts as the central processing unit. It receives input from the sensors, processes the data, and sends commands to the motors. This allows the robot to operate fully autonomously without human intervention during the competition.
- Other Components (Power System):
  The robot is powered by a rechargeable battery integrated into the controller. This battery supplies energy to all electronic components, including motors and sensors. Maintaining a fully

charged battery is important to ensure consistent performance and prevent loss of power during operation.

Mobility & Mechanical Design

The robot's mobility system is designed to ensure stability, control, and efficient navigation throughout the course.

- Design Description:

  The robot is built on a four-wheeled chassis using LEGO Spike components. The structure is designed to be rigid and balanced, keeping the center of mass stable during movement. The vehicle uses a car-like steering system, where the front wheels are responsible for changing direction while the rear wheels provide propulsion. This design improves control when navigating turns and reduces the likelihood of drifting off the track.

- Testing and Improvements:

  During development, the team conducted multiple tests using a replica of the competition field to simulate real conditions. The robot was repeatedly repositioned on the track to evaluate its consistency and accuracy from different starting points. Through testing, adjustments were made to wheel alignment, weight distribution, and sensor positioning to improve stability and navigation precision. These improvements resulted in smoother turns, better obstacle handling, and more reliable overall performance.

Power & Sensor Architecture

The robot's power and sensing systems are designed to provide stable performance and accurate environmental detection.

- Battery:

  The robot is powered by a rechargeable battery integrated into the control hub. This battery supplies consistent voltage to all components, including motors and sensors. Proper battery management is important, as low power levels can negatively affect motor performance and sensor accuracy, so we always keep it charged.

- Wiring System:

  All electronic components are connected to ensure reliable communication between the controller, sensors, and motors. Care was taken to secure the wiring to prevent disconnections or interference during movement.

- Sensor Placement:

  The sensors are mounted at the front of the robot to allow early detection of track markings and obstacles. Their position was chosen to maximize accuracy and provide sufficient reaction time for navigation adjustments.

- Calibration:

  Sensor calibration was performed during testing to improve accuracy and consistency. Threshold values were adjusted to ensure the robot

could reliably distinguish between different colors and detect obstacles under varying conditions.

Software Architecture & Obstacle Strategy

The robot operates using the LEGO Spike Prime program, and it continuously reads sensor inputs and adjusts its behavior in real time. The main logic is based on a loop structure where the robot:

1. Reads sensor data
2. Determines its position relative to the track
3. Adjusts steering and speed accordingly

For lane following, the robot uses color sensor data to detect boundaries and maintain its position within the track. For obstacle avoidance, the robot detects colored traffic signs and determines whether to pass on the left or right side, depending on the signal. The program then adjusts the steering angle to safely navigate around the obstacle. The code is structured into logical sections for sensor processing, decision-making, and motor control, allowing easier debugging and improvements.

Testing & Iteration

Problems we initially encountered when we first began testing were that the robot showed inconsistent behavior due to sensor inaccuracies and timing issues in the code.

The testing process involved multiple iterations to improve performance and reliability. The team conducted many structured test runs, adjusting sensor calibration and movement parameters after each attempt. Testing was performed on a replica of the competition field to simulate real conditions.

As a result, the robot achieved improved consistency in obstacle detection and smoother navigation.

Key improvements included removing unnecessary code elements and optimizing motor response time.

Systems Thinking & Engineering Decisions

The robot was designed with several **constraints,** including limited sensor accuracy, battery life, and mechanical stability. These constraints required careful balancing between performance and reliability.

A key tradeoff was reducing speed to improve accuracy. While this increased completion time slightly, it significantly improved success rates in completing the course.

Potential risks included loss of precision due to battery depletion and misalignment caused by repeated use. To address this, the team implemented regular calibration and ensured the battery was fully charged before each run.

Reproducibility

To reproduce this robot, a builder would need LEGO Spike Prime components, including motors, sensors, and structural pieces. The construction process involves assembling the chassis, attaching the motors to the drive system, and mounting the sensors at the front of the robot. After assembling the hardware, the control program must be uploaded to the controller. The program handles sensor input, movement control, and obstacle navigation. Calibration is required to adjust sensor sensitivity and ensure accurate performance. This includes testing the robot on a track and adjusting values as needed.

- Build steps: ++++++++++++++++++++++++++++

- GitHub address: https://github.com/PRVRobotics/Sojourner-WRO-FutureEngineers-2026/tree/main

Future Improvements

In the future, we plan to improve the robot’s programming to make it more reliable and fully autonomous. This would allow the robot to respond more effectively to different challenges with greater consistency. We also aim to optimize the control logic to improve decision-making and reduce delays in response time.

We plan to enhance the wheel design to improve overall stability and movement precision. Improving wheel alignment and traction would result in smoother navigation and better control, especially during turns. Additionally, reinforcing the wheel's attachments would reduce the risk of mechanical failure during operation.

Another area for improvement is the sensor system. We could refine sensor placement and calibration to increase accuracy in detecting track lines and obstacles. Adding more advanced sensing or improving how data is processed could help the robot react faster and more precisely.

We would also like to improve the overall structure of the robot by optimizing weight distribution and strengthening key components. This would increase durability and help maintain consistent performance over time.

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